

Lynn Margulis, 1938–2011

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Unquestionably the world's leading advocate of the role of symbiosis in evolution, Lynn Margulis died at her home in Amherst, MA on November 22, 2011 after suffering a stroke. She was Distinguished University Professor of Geosciences at the University of Massachusetts Amherst and a member of the National Academy of Sciences of the USA. Furthermore, Lynn was a recipient of the National Medal of Science, the highest award in the USA for a scientist, and of the Linnean Society of London's Darwin-Wallace medal. She leaves three sons, Dorion, Jeremy and Zachary, and a daughter, Jennifer.

Lynn Margulis was born in Chicago, and attended the University of Chicago Laboratory School. She received her Master's degree in genetics and zoology at the University of Wisconsin in 1960 and her Ph.D. in genetics from the University of California, Berkeley in 1963. She taught briefly at Brandeis University and then for 21 years at Boston University before moving to the University of Massachusetts Amherst in 1989. She never retired, and continued teaching her signature course, Environmental Evolution, twice a year.

A tireless advocate for what became known as the Serial Endosymbiosis Theory of Cell Evolution, she advanced the cause of science not by her primary discoveries, but through her synthesis of many peoples' work, first in her classic paper, "On the origin of mitosing cells" (1967 as Lynn Sagan), then in the publication of *Origin of Eukaryotic Cells*

(1970) and finally in her *magnum opus* *Symbiosis in Cell Evolution* (1981, 1993). She had just completed a new edition of this work, to be published under the title, *Symbiogenetics*, prior to her death. These books were groundbreaking in that they incorporated geological data into her hypotheses on cell evolution, combining paleontology and microbiology, and presaging many biologists' study of early life (See Archibald 2011) (Fig. 1).

She was heavily influenced by Wisconsin Professor Hans Ris, who with Walter Plaut, had been the first to use electron microscopy to show that chloroplasts had bacteria-type DNA, and by Professor Elso Barghorn of Harvard University. Barghorn and Tyler had shown with their study of the Gunflint Chert that life on Earth was much older than previously thought. While teaching at Boston University, Lynn was a frequent visitor of Barghorn's laboratory, where she learned of microbial mats and stromatolites. She would go on to influence NASA's interest in these microbial communities that served as analogs of early life on Earth. Barghorn's influence on her work can be seen in her 1967 *Journal of Theoretical Biology* paper, and, in her 1970 book *Origin of Eukaryotic Cells*.

In the early 1970s Lynn was active in NASA's origin of life meetings, editing several of the volumes. She also served on the influential Space Science Board, and was a leader in the NASA Exobiology program for many years, as well as in the Marine Biological Laboratory-based NASA Planetary Biology Internship program, which she directed for over 30 years. Her work for NASA is described in S.J. Dick and J.E. Strick's excellent 2004 book *The Living Universe: NASA and the Development of Exobiology*.

Lynn went on to collaborate with scientists in many fields, including James Lovelock, Heinz Lowenstam, William Nutting, and many others. She delighted in bringing people together, introducing researchers to each other, and cross-fertilizing the different disciplines. In recent years Lynn's view of evolution also attracted the attention of social scientists, including Canadian sociologist Myra Hird, whose 2009 *The*

The upcoming International Symbiosis Congress July 22–28 in Kraków, where she had been scheduled to give a keynote presentation, will be dedicated in memory and tribute to her.

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Origins of Sociable Life: Evolution After Science Studies, was written during a year's sabbatical in the Margulis lab. The book explored, among other things, how concepts of symbiosis could be applied to social relations.

While not a taxonomist, per se, Lynn played a major role in compiling taxonomic information in her influential *Five Kingdoms* (with Karlene V. Schwartz), which was recently released in a fourth edition under the title *Kingdoms and Domains* (co-authored with Michael Chapman). In his preface to the first edition, Stephen Jay Gould wrote "Margulis and Schwartz have generated here that rarest of intellectual treasures – something truly original and useful." Margulis was particularly interested in protists, or protoctists, as she insisted on calling them, and produced the monumental *Handbook of Protoctista*, and an accompanying *Illustrated Glossary*, that she modeled after *The Prokaryotes*. A second edition of the *Handbook* will be published this year. While never joining the Society of Protozoologists (now the International Society of Protistologists), she was a founder, with Max Taylor and Howard Whisler, of the International Society of Evolutionary Protistology.

One can't recall Lynn without reflecting on her signature hypothesis – that the eukaryotic flagellum, or undulipodium, evolved symbiogenetically from spirochete bacteria. She developed the idea after reading the following comment in Herbert F. Copeland's 1956 *The Classification of Lower Organisms*: "Spirochaets which share the habitat of these [termite gut flagellates] are commonly found adhering to their cell membranes, and were mistaken for additional flagella in the original descriptions of some of the genera." While the termite gut microbial community presented many intriguing questions regarding the relationship between the myriad of spirochetes in the gut and the symbiotic flagellates, no molecular or biochemical data were ever developed to support the idea that these motility symbionts evolved into cell organelles. In recent years she had focused more on the question of spirochete genetic contributions to the eukaryotic genome, developing an interest

in Galina Dubinina's *Thiodendron* consortium, (the spirochete of which has recently been described as *Spirochaeta perfilievii*), and sponsoring the work of John L. Hall on this question leading to his recent publication in *Symbiosis* (Hall 2011).

Lynn knew that one of the ways to have symbiosis become recognized for its central impact on life systems was to help initiate and promote an international group of symbiosis researchers. Thus, her role in helping establish symbiosis conferences and then co-founding the International Symbiosis Society 15 years ago, was critically important. She presented talks and films at nearly every ISS Congress including a keynote address at the Halifax meeting. With the growing interest in the symbiosis fields over the past two decades, it often appeared that Lynn's dynamic and deeply thought-provoking examples and tireless advocacy of symbiosis were a consistent fuel for countless important research efforts in the field that continue to this day.

Lynn's influence on teachers was profound and unique as well. Few scientists in American history can claim such a direct and consistent commitment to education at all levels. One multi-year example perhaps suffices to illustrate this. Lynn helped to initiate the International Microcosmos Project, established at Boston University's School of Education Science Education Department (one of us, D. Zook, as Director). Established initially, simply with creative teacher and student input, a curriculum emerged that showed how microbes can be a powerful learning vehicle in pre-college classrooms. Later the project was funded by the National Science Foundation and over a 10 year period, more than 450 workshops (some over 90 contact hours!) were conducted, reaching thousands of teachers in nearly all the US States and six other countries. Although Lynn was at the University of Massachusetts during much of this time, she frequently took time from an impossibly busy schedule to lead interactive sessions within the workshops on foraminiferans, microbes as primary recyclers, and in symbiotic systems. Besides her many visits which inspired the teachers, she helped to arrange for leading science colleagues such as Dr. Ricardo Guerrero from Autonomous University in Barcelona to come and work with the Microcosmos teachers. She once noted to me (Zook), "...Our findings and ideas in symbiosis, really in science as a whole, are most meaningful when we can bring it to teachers and their students..."

Forthcoming conferences, meetings, papers, publications will give tribute to Lynn and all that she contributed to the world of science and scientific thinking. It will take all these words, pronouncements, and published pages or proceedings to reflect Lynn Margulis's immense impact. As one symbiosis research colleague put it, "It's as if her absence

is a tear in the very fabric of the universe.” She truly inspired a generation of scientists and will continue to inspire us all.

Lynn’s old friend Eugene Garfield has constructed an on-line bibliography of over 250 of her articles that can be found at this link

<http://garfield.library.upenn.edu/histcomp/index-margulis.html>

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